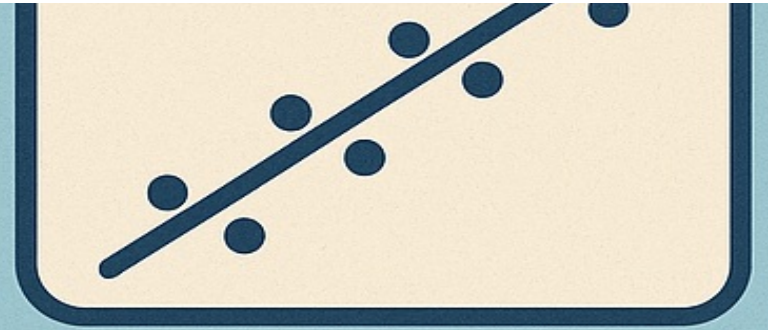
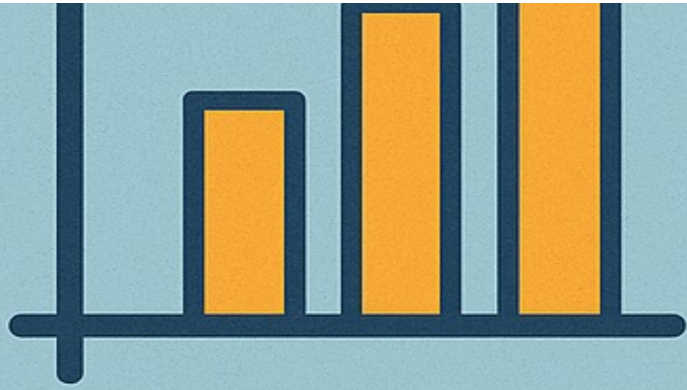
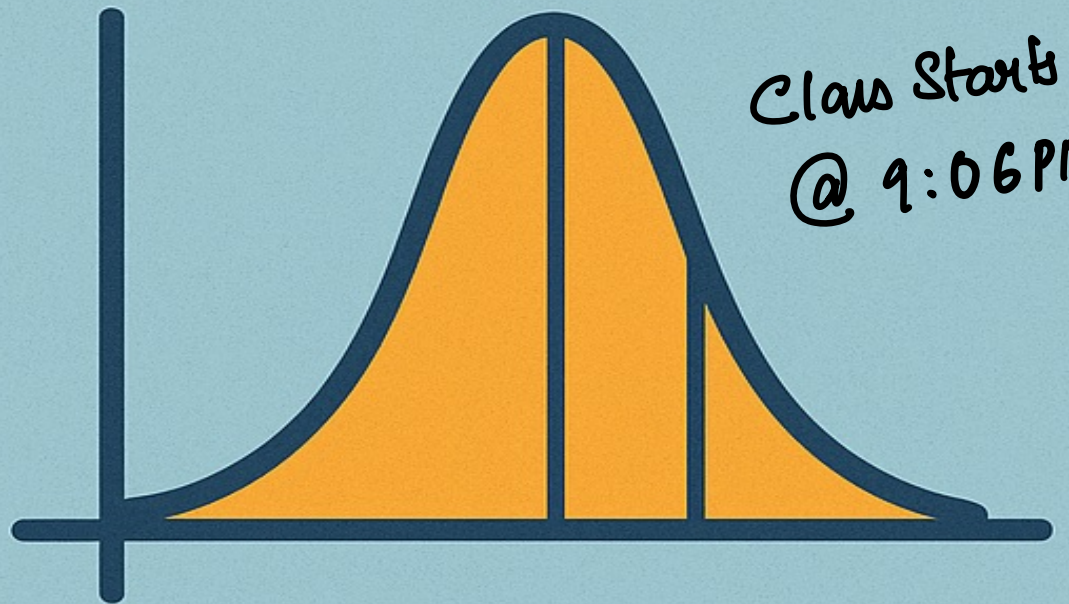
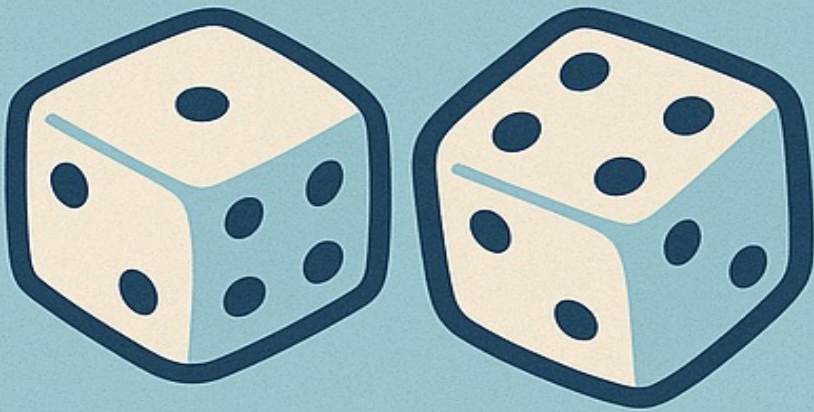




PROBABILITY AND STATISTICS



Class Starts
@ 9:06 PM

Advanced Probability Distributions

Agenda

1) Confidence Interval with Bootstrapping $\rightarrow$ CLI

2) Poisson distribution ✓

3) Exponential distribution ✓

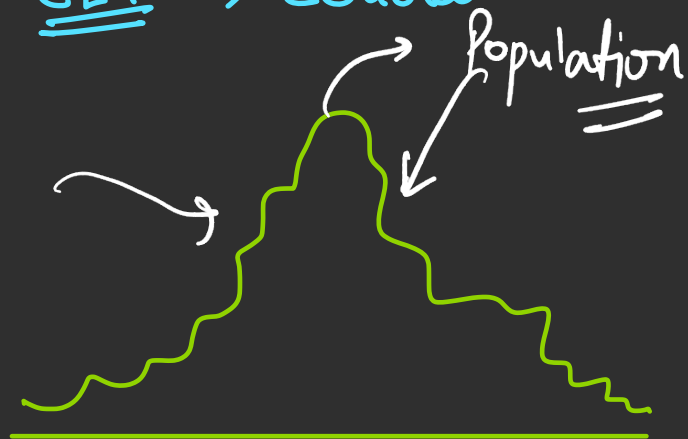
4) Geometric distribution ✓

$\rightarrow$ Wed' session

Recap

$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ Drawback of CLT? Point estimate

1/ CLT → Central limit theorem



Not Normal

$(\mu, \sigma, \text{var})$

$P(X \leq 69.1)$

CDF

$P(65.1 < X < 70.1) \rightarrow P(Z)$

sample $(n \geq 30)$

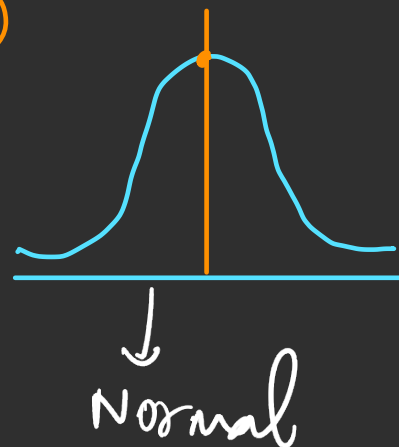
sample size (n)

Calculate the mean of that sample

(1000 times repeat)

1000 sample mean

plot this mean



(Means of the sample means $\hat{=}$ popⁿ mean)

Small law $\rightarrow$ point estimate

CLT

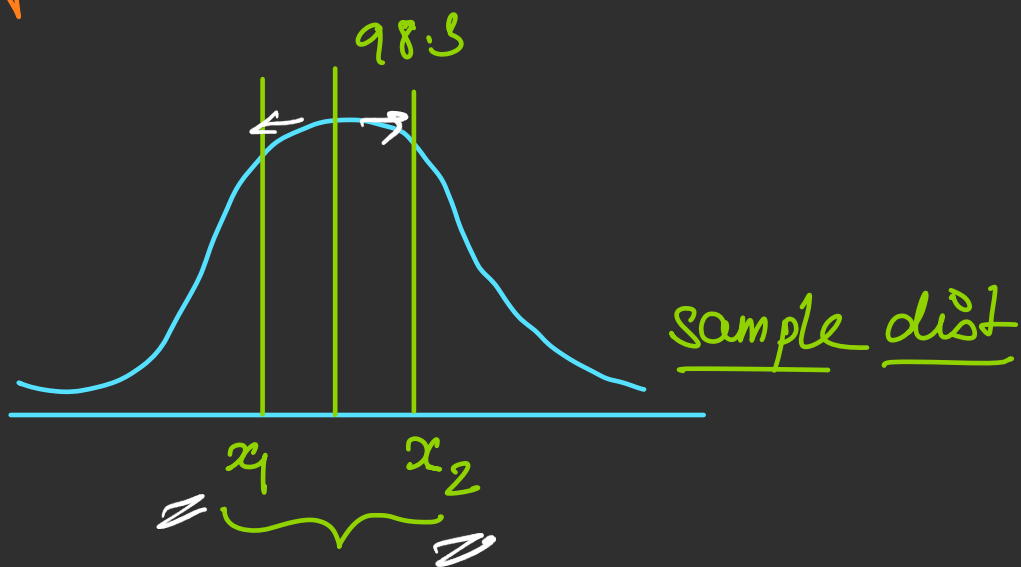
$\hookrightarrow$ mean

$\hookrightarrow$ Confidence Interval

(The range in which the popⁿ parameter lies)

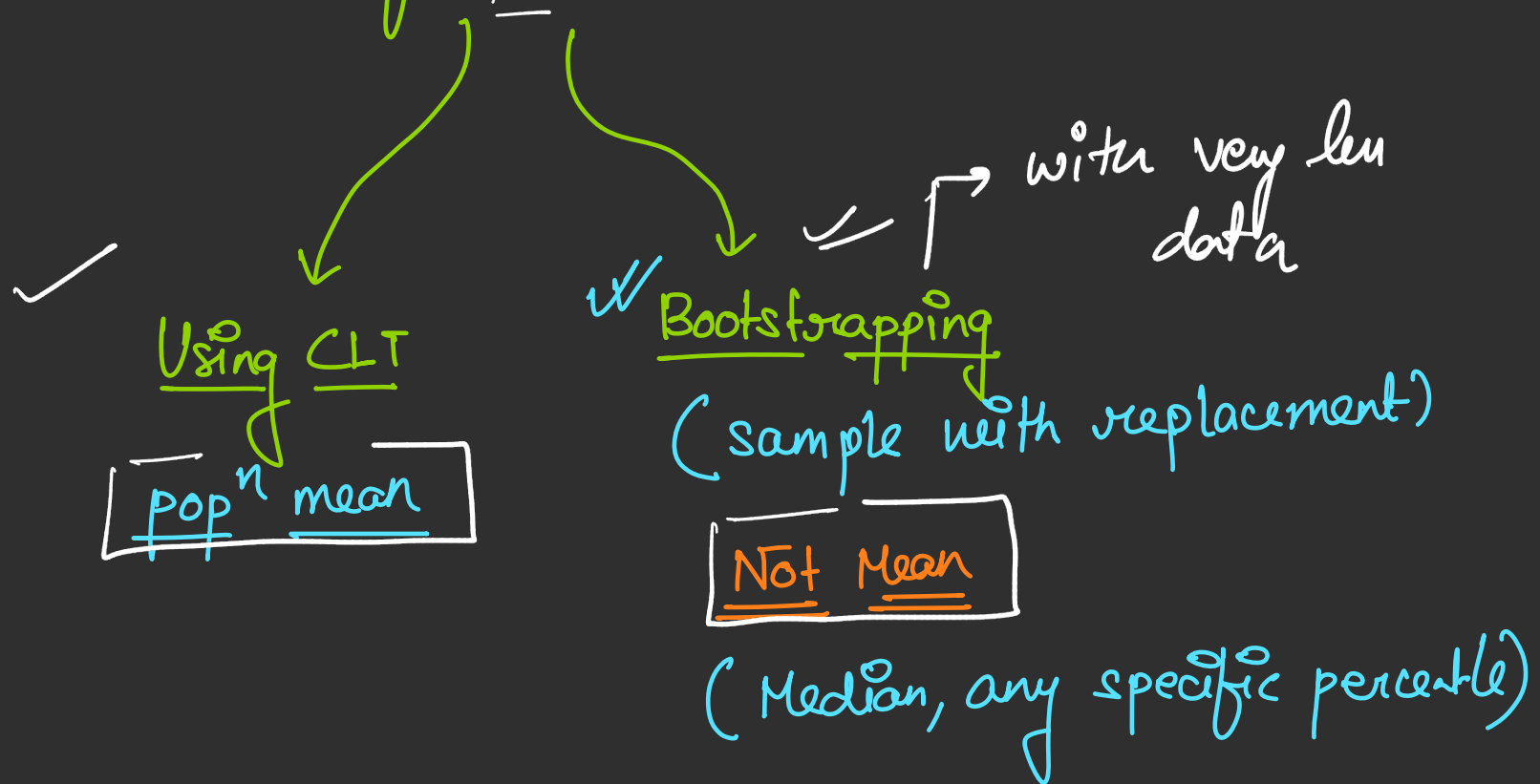
CLT

$\hookrightarrow$ (98.3 $\hat{=}$ popⁿ mean)



$$\begin{aligned}
 \text{CI} &= \underbrace{\text{point estimate}}_{\substack{\text{pop}^n \text{ mean} / \\ \text{sample mean} \\ \text{(Using CLT)}}} \pm \underbrace{\text{Margin of Error}}_{\substack{Z_{\text{value}} \times SE \\ \left(Z \times \frac{\sigma}{\sqrt{n}} \right)}}
 \end{aligned}$$

Confidence Interval



[This method is useful when you have very small dataset]

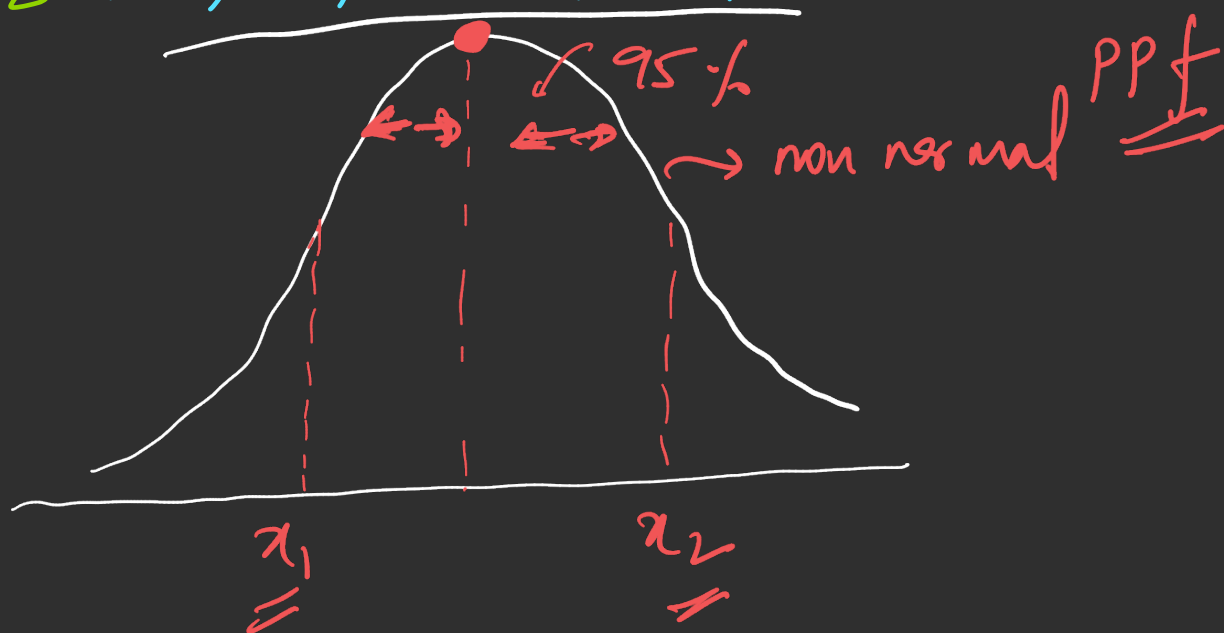
Salary data

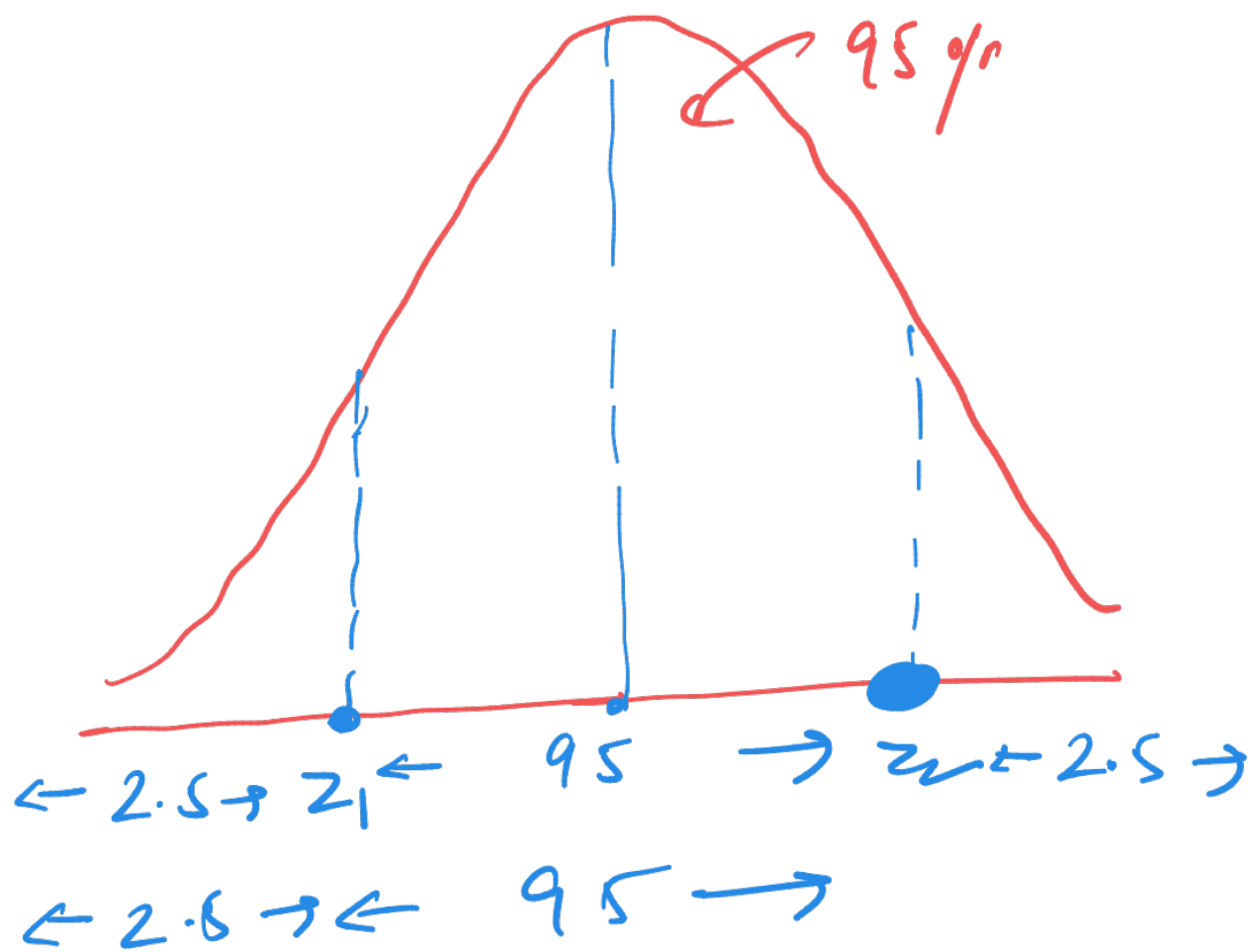
✓

$$D_1 = [35, 36, 33, 37, 34, 35] \rightarrow \underline{\underline{35L}} \quad \text{Mean}$$

✓

$$D_2 = [20, 37, 17, 50, 53, 33] \rightarrow \underline{\underline{35L}}$$





5%

100 - 5%

= 95

97.5

n independent trials meaning
probability of success / failure
↳ binomial

bernoulli

← 1 trial only prob of success / failure

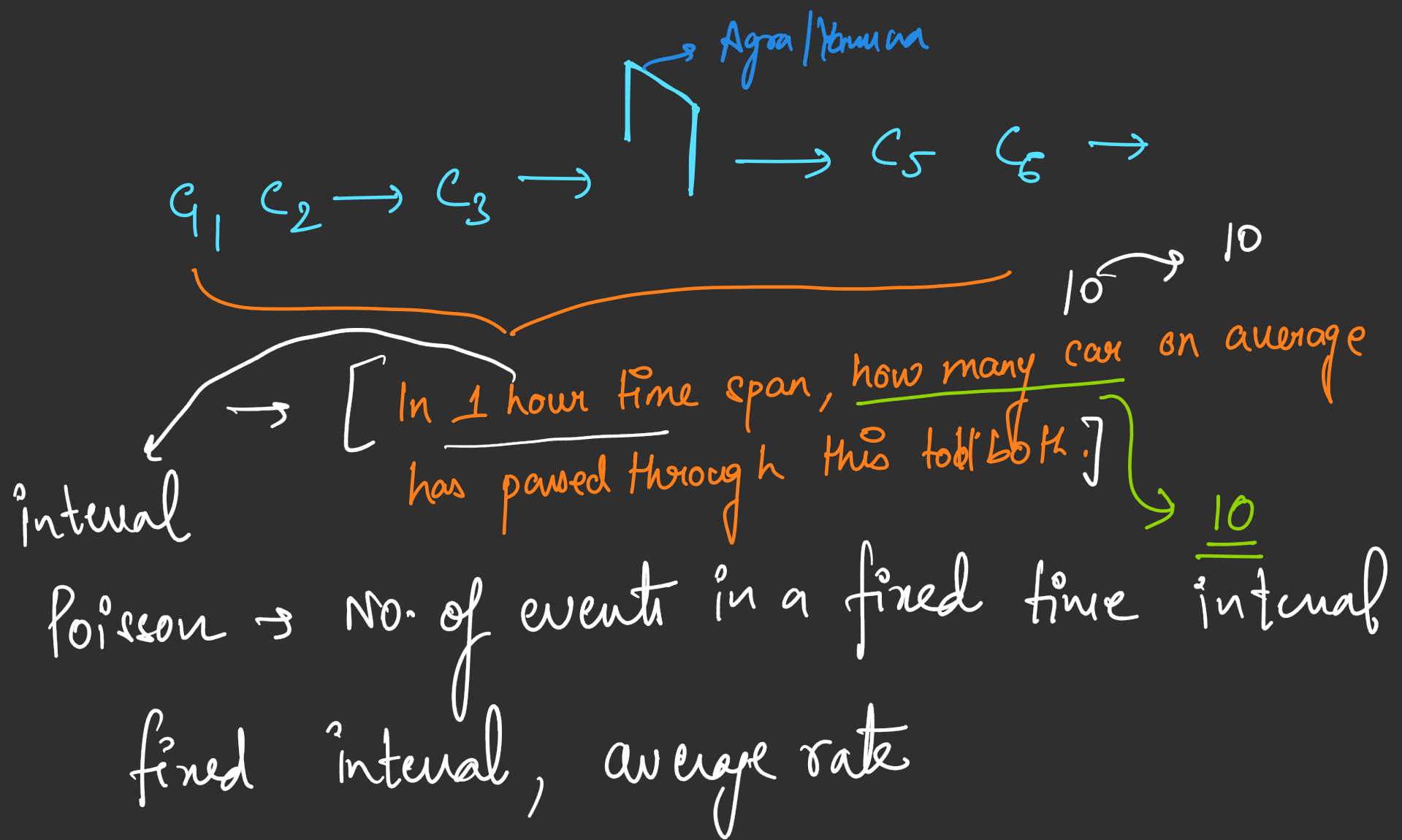
which is the disto where the outcomes are
equally

↳ uniform distribution

↳ both

Normal → always continuous

Poisson Distribution



Model the number of events that occurs in a fixed interval of time & space.

PD → Discrete distribution

→ Fixed interval → 1 hour

→ Average rate (λ) → 10 car/hour

Q/ What is the prob of exactly 25 vehicle passing through the toll booth in next 1 hour?

Formula

$$P[X = \underline{k}] = \frac{\lambda^k * e^{-\lambda}}{k!}$$

25

$$10^{25} * e^{-10} \\ \hline 25!$$

$\lambda \rightarrow$ Average rate in a specific interval

$\underline{e} \rightarrow 2.71$

$\underline{k} \rightarrow$ No of events

Poisson Approx to Binomial dist

There are 80 students in a kinder garden class.
Each one of them has 0.015 probability of forgetting their lunch on any given day.

(a) What is the average or expected number of students who forgot lunch in the class?

(b) What is the probability that exactly 3 of them will forget their lunch today?

$$\text{Avg} / E(x) = \underbrace{80 \times 0.015}_{\text{total state}} \quad (\text{lambda})$$

$$P[X=3] = \text{poisson.pmf}(x=3, \mu=1.2)$$

$\Rightarrow$

$$\text{Avg state} = \underbrace{n}_{80} \times \underbrace{p}_{0.015}$$

Exponential Dist

Example: Q. You receive 240 messages per hour on average - assume Poisson distributed. Rate of messages arriving per second is $\frac{1}{15}$.

Q1) What is the probability of having no message in 10 seconds?

$$P[X=k] = \frac{e^{-\lambda} \times \lambda^k}{k!}$$

$k \geq 0$

$\frac{10}{15} = 10 \times \frac{1}{15}$

λ

$1 - \frac{1}{15} = \frac{10}{15}$

$$\frac{e^{-(10/15)} \times \lambda^0}{0!} \rightarrow 1$$

$$\Rightarrow \boxed{e^{-(10/15)}} \Rightarrow \underline{\underline{e^{-10\lambda}}}$$

$$e^{-(10 \times \frac{1}{15})} \Rightarrow \underline{\underline{e^{-(10\lambda)}}}$$

Probability 0 messages

$$= \frac{\lambda^k x e^{-\lambda}}{k!}$$

$$= \frac{e^{-\lambda} x \lambda^0}{0!}$$

$$= e^{-\lambda}$$

$$= e^{-10/15}$$

Formula

$$P[T > 10] = e^{-\lambda x}$$

$\lambda = \text{rate}$

$$P(T \leq x) = 1 - e^{-\lambda x}$$

$\downarrow$ cdf

cdf

$$P[T > 10] = e^{-\lambda x}$$

$$P[T \leq 10] = 1 - e^{-\lambda x}$$

$$\text{rate} = \lambda$$

Exp dist

$$(\beta)$$

$$\beta = 1/\lambda$$

scale parameter

$$P(x, \lambda) = \lambda e^{-\lambda x}$$

pdf

$$P[T > 10]$$

$$\rightarrow \text{scale} = \frac{1}{\text{lamba}} \Rightarrow \frac{1}{(1/15)} = \underline{\underline{15}}$$

$$1 - \text{exp.cdf}(x=10, \text{scale}=15)$$

$$SE_1 + SE_2 \Rightarrow \frac{\sigma}{\sqrt{n_1}} + \frac{\sigma}{\sqrt{n_2}}$$

$$\underbrace{x_1 - x_2}_{-5} =$$

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

✓

✓

$$N\left(\underbrace{\bar{x}}_{\checkmark}, \underbrace{\frac{\sigma}{\sqrt{n}}}_{\checkmark}\right)$$

SE

$$P(\underbrace{x_1 - x_2}_{\text{wavy}} > \underline{0})$$

$$\rightarrow z = \frac{0 - (-5)}{4.88} = \underline{\underline{1.024}}$$

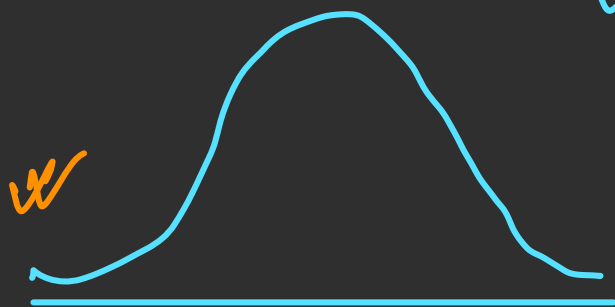
$$P(\underbrace{z > 0}_{\text{wavy}})$$

$$\rightarrow 1 - \text{norm.cdf}(\underline{\underline{1.024}})$$

$$\checkmark P(\checkmark \underbrace{\bar{x}_1 - \bar{x}_2}_{\text{wavy}} > 0)$$

$$\bar{x}_1 - \bar{x}_2 = \underline{\underline{-5}}$$

$$\sim \underline{\underline{N(-5, 4.08)}}$$



$$\left(\underbrace{\bar{x}_1 - \bar{x}_2}_{-5}, \underbrace{SE_1 + SE_2}_{4.08} \right)$$

$$Z = \frac{x - \bar{x}}{SE} = \frac{0 - (-5)}{4.08}$$

$$\Rightarrow \underline{\underline{1.023}}$$

$$\underline{\underline{1 - cdf(z)}}$$